

Business Introductions

Microsoft Corporation is a global technology company that develops and sells software, cloud services, devices, and digital solutions to businesses, consumers, and governments worldwide. Based on Microsoft's revenues (via their income statement), the company makes the most money in these two ways: Product revenue - this is revenue from selling software licenses and physical/digital products. These may include Xbox hardware, Windows licenses, Surface devices, or Server Licenses. These products are mainly one-time purchase and ownership based, and see about $\frac{1}{3}$ of the total revenue for Microsoft. The second and biggest way Microsoft receives revenue is through services. This includes Azure (cloud service), Microsoft 365 subscriptions, LinkedIn subscriptions and ads, github subscriptions, xbox game pass, support services and consulting, and other advertising on bing and linkedin. Through services and other revenue, Microsoft was able to establish a recurring, usage based revenue stream, that makes up about $\frac{2}{3}$ of their total revenue in FY2025.

Microsoft has seen consistent success year after year, whether that's in their revenue, gross profit, or operating income - the company has been able to sell more and more products and services reliably to establish itself as a tech giant. However, why has the stock slid in the past year?

This paper will dive into why companies (like Microsoft), despite having impressive numbers, still fail to meet investor elevated expectations the market had priced in, and how free cash flow is one of the most important factors to look at when valuing a company.

The Current Situation

If Microsoft consistently profits off of their recurring revenue streams, then why is the stock down 3.66% over a year when the business grew? To start off, let's review the company's financial statements: In the income statement, Microsoft has repeatedly reported consistent revenue growth, as the company went from \$198.27B in FY2022 to \$281.72B in FY2025 (42% revenue growth in 3 years). However, in the cash flow statement Microsoft's capex has grown from \$-23.89B in FY2022 to \$-64.55B in FY2025, reflecting the data center investments. Next, let's speak about the company's gross, operating, and net margins to back up the businesses growth: From FY2022 to FY2025, gross margin expanded from 68.4% in FY2022 to 68.8% in FY2025 - a 0.4% improvement over the full four year period. Even though the gross margin has remained stable from FY2022 to FY2025, the company being able to generate nearly 70 cents of every revenue dollar after direct costs is extraordinary, compared to Apple's gross margin of 46.9% in FY2025. This shows that Microsoft's structure is sufficient to maintain large revenue streams, as their software and cloud economics are superior to Apples. This is why when companies create software, they are able to consistently profit off of them without having to spend too much money YoY. Secondly, from FY2022 to FY2025, Microsoft has improved their operating margin from 42.1% to 45.6%, showing that Microsoft is able to efficiently run their business while investing heavily. Revenue is growing faster than operating expenses. It is also worth mentioning that Microsoft has a net margin of 36.1%, meaning that the company keeps 36 cents of every dollar it earns as profit. This is the one of the highest net margins of any large company on earth. Even through all of this growth, the main reason why the stock tumbled was because of one ratio ~ the FCF conversion. The free cash flow conversion dropped from 0.90x to 0.70x, and is purely because of the companies spending on equipment to prepare for AI development. This can be seen through the Capital Expenditure changes throughout the year, where CapEx went from \$28.11B in FY2023 to \$44.48B in FY2024, and finally \$64.55B in FY2025. Microsoft converting its operating cash into physical infrastructure rather than letting it sit as free cash is a risky decision - especially if they do not impress investors in their next earning reports.

At any angle, these numbers reflect a strong, reliable, and structured business. But still, why does the market seem to be bearish as the stock has slid 3.66% over a year? The answer lies deeply in the capex spending. Even though Microsoft's revenue, gross, and operating growth has been consistent, investors are still unconvinced that the company's increasing capex is paying off as it should. Last year, investors bet that Microsoft would see higher numbers in their growth given capex spending - which hasn't been the case. This is the sole reason why the company has slid 3.66% over the last year; investors aren't seeing the progress they want to see from a company spending more and more free cash every year.

The valuation/Discounted Cash Flow

Based on Microsoft's business structure (a structure of reliability), P/E, P/S, and EV/EBITDA reflect that. In FY2025, Microsoft's P/E sat at around 27.2x - which is cheaper than tech giant Apple (33.1x) despite having superior margins and faster revenue growth. At 17.5x EV/EBITDA, Microsoft is cheaper than Apple's 26.1x. However, Microsoft's P/S of 9.8x means that the company demands a high premium to be a part of its fast growing business. Even though Microsoft has reported impressive growth over the past few years, the market is still expecting more from the company, and we can see this through the DCF.

The Discounted Cash Flow answers one question: if I bought the entire Microsoft business today, how long would it take to get my money back from the cash it generates, and what is that cash worth in today's dollars? Let's first start off by taking the 4 year average of the free cash flow of Microsoft: \$65.15B (FY2022) + \$59.48B (FY2023) + \$74.07B (FY2024) + \$71.61B (FY2025) = \$67.58 B. This is our starting point to avoid irregularities in FY2025 in case something dramatic happened (reflects the company better). Now, Microsoft's FCF has been growing around 3.2% per year on average historically, so we multiply the average FCF from the last 4 years by 3.2% and do this for the next 5 years. Next, we need to calculate the PV (present value) of that future cash flow. This is where the discounting happens. The formula for this is $PV = FCF / (1 + WACC)^{(year\ number)}$. With this, we get these numbers respectively:

Year 1: \$69.74B → PV \$63.98B

Year 2: \$71.97B → PV \$60.58B

Year 3: \$74.28B → PV \$57.36B

Year 4: \$76.66B → PV \$54.31B

Year 5: \$79.11B → PV \$51.42B

However, after the 5th year, the model doesn't just stop. Microsoft will most likely keep generating cash forever, so we need to account for that. Terminal value captures all of that infinite future cash in one number using this formula:

$$\begin{aligned} \text{Terminal Value} &= \text{Year 5 FCF} * (1 + \text{terminal growth rate}) / (\text{WACC} - \text{terminal growth rate}) \\ &= \$79.11B * 1.03 / (0.09 - 0.03) \\ &= \$81.48B / 0.06 \\ &= \$1,358B \end{aligned}$$

Last step - from enterprise value to share price:

Enterprise Value: \$1,170B (sum of all discounted cash flows) – Total Debt: \$60.6B (debt holders get paid before equity holders) + Cash: \$30.2B (cash on hand belongs to shareholders) = Equity Value: \$1,140B (what belongs to shareholders) ÷ Shares: 7.43B (split equally among all shares) = Implied Price: \$153.52

The implied price of \$153.52 versus the current price of \$356.77 looks alarming at first, but the model is using 3.2% historical free cash flow compound annual growth rate (FCF CAGR) which understates Microsoft's power trajectory. FCF is being suppressed by the \$64.6B in annual capex, and operating cash flow grew 53% in 3 years; if capex stabilizes, FCF will surge. If the data center buildout is largely complete in 2-3 years and capex normalizes back towards \$35-40B, then FCF would be approximately:

Operating Cash Flow FY2025: \$136B

Normalised Capex: \$40B

Normalised FCF: \$96B

A more realistic valuation of Microsoft accounts for normalized capex. The historical DCF understates Microsoft's true earnings power because the \$64.6B capex surge is temporarily compressing FCF. Once data center buildout stabilizes and capex normalizes toward \$35-40B, FCF would jump from \$67B four-year average to approximately \$96B - pushing the DCF much closer to the current market price. The factor that is difficult for Microsoft to work through is time. Time involves steadiness and patience, even when investors think that something is going wrong. For Microsoft, as long as they don't continue increasing capex indefinitely, then that should reflect well in their financial statements in the next few years. That \$96B FCF versus the \$67B four-year average the model used would likely put the DCF's implied stock price at around \$250-300.

Conclusion

Given the financial statements, the financial multiples, and the DCF, I believe that Microsoft is fairly valued if you believe AI capex investment will pay off within the next 2-3 years, and overvalued if it doesn't. The DCF using the historical FCF instead of normalized FCF (once capex stabilizes), undervalues the company. In response to this, the market has been selling shares of Microsoft because of their unimpressive earnings reports given their heavy investments in AI. Even though the company has been consistently making money, the company is investing more and more money into AI, without showing the benefits through their earnings reports. I believe that in the future when the company reduces their capex, the market will reward Microsoft because of this.